Matteo **Bruno**

Associate Researcher at Sony CSL

About me

I am currently working as a researcher at Sony CSL in Rome, focusing on urban mobility and sustainable cities. I also teach a course about mathematical modelling in urban planning at the faculty of Architecture in Roma Tre University.

As a trained mathematician and a PhD in complex systems, I love trying to solve challenging problems. I also like an open and stimulating work environment, which I take as an opportunity for studying and discovering new things in every subject. I am generally very curious and I like to apply scientific methods to better understand and improve the world, searching for out-of-the-box solutions. However, I am a pragmatic person and love working with a goal in mind.

I am always open to new enthralling projects with the purpose of improving the well-being of the planet.

Experience

Sony CSL — *Rome, Italy*

Associate Researcher — *Oct. 2021 - Present*

- Working in the sustainable cities team headed by Professor Vittorio Loreto.
- My research is currently focusing on urban mobility and sustainability.
- Supervising PhD and master students.
- Working on European projects: S+T+ARTS: Repairing the present, S+T+ARTS: Air, S+T+ARTS: Buen-Tek, DUT Driving Urban Transitions: PROWD

University of Oxford — *Oxford, UK*

Visiting fellow — *Feb. 2020 - Apr. 2020*

- Supervised by Professor Renaud Lambiotte at the Faculty of Mathematics.
- Working on Twitter discussion during Brexit.

University of Bogotá Jorge Tadeo Lozano — *Bogotá, Colombia*

Visiting fellow — *Oct. 2018 - Dec. 2018*

- Supervised by Professor Marco Dueñas.
- Analysing Colombian firms' exports activities.
- Learned Spanish before and during the visiting period.

Teaching experience

2023-2025	**Mathematical and statistical models and methods**, Part of the course *Urban planning laboratory: re-inhabiting the urban*, faculty of Architecture	*Roma Tre University*
2021-2024	**Selected lectures at the Complex Systems course**, City modelling, faculty of Physics	*Sapienza University*
2017	**Introduction to mathematics**, Introductory course of general mathematics	*UNINT*

Education

IMT School for advanced studies Lucca — *Lucca, Italy*

PhD in Systems Science — *Sep. 2017 - Oct. 2021*

- PhD thesis titled *Maximum entropy methods for the statistical analysis of bipartite networks: fast computation and applications*.
- Worked in the NETWORKS research unit under the supervision of Prof. Diego Garlaschelli, Prof. Guido Caldarelli and Dr. Fabio Saracco.
- The focus of my work at IMT has been on models and application of bipartite networks to social, ecological and economic systems.

Roma Tre University — *Rome, Italy*

Master Degree in Mathematics — *Sep. 2015 - Jul. 2017*

- 110/110 cum laude with a thesis titled *Weak community detection in the stochastic block model and its phase transition* under the supervision of Professor Pietro Caputo.

University of Edinburgh — *Edinburgh, UK*

Visiting student, Erasmus+ scholarship — *Sep. 2014 - Dec. 2014*

Roma Tre University — *Rome, Italy*

Bachelor Degree in Mathematics — *Sep. 2012 - Jul. 2015*

- 110/110 cum laude.

Skills

Languages spoken Italian (native), English (fluent), Spanish (proficient)
Programming skills Python, QGIS, Matlab, JAVA, C, LaTeX, Mathematica
Methodologies expertise Geospatial data science, Data science, Data visualization, Mathematical modelling, Network analysis, Machine learning

Extracurricular Activity

IMT Lucca *Lucca, Italy*
STUDENTS' REPRESENTATIVE *Nov. 2017 - OCt. 2020*

- Representative of the students of the XXXIII PhD cycle at IMT Lucca.

Complexity72H *Lucca, Italy*
PARTICIPANT *June 2019*

- Participant in the workshop Complexity 72h. In this workshop, the participants were divided in groups and supervised by a professor to write a publishable paper in 72 hours.
- Supervised by professor Marián Boguñá, me and my team worked on the paper Community Detection in the Hyperbolic Space.

Soccer data challenge *Pisa, Italy*
PARTICIPANT *September 2018*

- Participant in the Soccer data challenge 2018. In this challenge, each team had to analyze datasets of movements and actions by players in Serie A football teams and solve football-related problems to create measures for the performance of players and teams.

INDAM Scholarship *Italy*
WINNER *September 2012*

- Winner of the INDAM (High mathematics Italian national institute) scholarship for the mathematics class of 2012. Classified 12th among all national applicants.

Soft skills

Problem Solving As a mathematician and a complex systems scientist, I am always attracted by new problems. I search not only for a solution, but I like to find fast, efficient and elegant solutions.

Team working I believe this skill is not only about the working moments but also about creating a friendly and inclusive environment. I have learned during my PhD to create and work in a team in a harmonious and efficient way. Living in an international campus for years and being representative of the students really enhanced my social skills in any situation.

Working under pressure During my PhD studies, pressure has been a part of the course of study and I have learned to handle deadlines and obstacles while managing multiple tasks at the same time.

Perseverance I can focus on a big project for a long time if I am motivated: as an example, I ran the Rome Marathon in 2014 with a time of approximately 4:40.

Packages managed

BiCM Python package for the analysis of bipartite networks. Package at github, documentation at Read the Docs.
NEMtropy Python package for the computation of maximum entropy network models. Package at github, documentation at Read the Docs.

Conferences and workshops (selected)

2024	**Participant**, Pisa, Italy	*European Mobility Symposium*
2024	**Organizer of the satellite event: *UrbanSys2024***, Exeter, UK	*UrbanSys2024*
2024	**Lightning talk: *A universal framework for inclusive 15-minute cities***, Exeter, UK	*CCS2024*
2024	**Presented the talk: *Exploring the feasibility of the 15-minute city by equalising local accessibility***, Venice, Italy	*NetSci-X*
2023	**Presented the talk: *Can my city become a 15-minute city? Designing equal accessibility***, Vienna, Austria	*NetSci*
2022	**Presented two talks: *Brexit and bots: characterizing the behaviour of automated accounts on Twitter during the UK election* and *Assessing the 15-Minute city and its impact on the quality of life***, Palma de Mallorca, Spain	*CCS2022*
2021	**Presented two talks: *The ambiguity of nestedness under soft and hard constraints* and *BiCM: a fast Python package for the computation of the maximum entropy bipartite configuration model and its statistical one-mode projection***, Online	*Networks 2021*
2020	**Presented the talk: *Thermodynamics disentangles conflicting notions of nestedness in ecological networks***, Rome, Italy (Online)	*NetSci 2020*
2020	**Participant**, Tokyo, Japan	*NetSci-X 2020*
2019	**Presented the talk: *An entropy-based approach for pattern detection on Twitter: exact vs approximated methods***, Lucca, Italy	*TOFFEE EU project*
2019	**Poster presentation: *Colombian export capabilities: building the firms-products network***, Trento, Italy	*CCS/Italy*
2018	**Presented the talk: *Colombian export capabilities: building the firms-products network***, Bogotá, Colombia	*Innovación, dinámica industrial y política industrial*

Talks

2025	**Invited speaker: *X and the city: quantifying urban livability through data***, L'Aquila, Italy	*Mathematics, University of L'Aquila*
2025	**Invited speaker: *Quantifying proximity and distant opportunities to design accessible cities***, Rome, Italy	*Cornell University Architecture Art Planning @Rome*
2024	**Invited speaker: *From the 15-Minute City to Distant Opportunities: How Inequalities of Accessibility Shape Our Cities***, New York, USA	*CUNY*

Publications

2025	**R.Basilone, M. Bruno, H.P.M. Melo, M. Avalle, V. Loreto. *Road-Width-Aware network optimisation for bike lane planning***, doi.org/10.1088/2632-072X/adf683	*Journal of Physics: Complexity*
2025	**F. Fanelli, H.P.M. Melo, M. Bruno, V. Loreto. *Revealing the core-periphery structure of cities***, doi.org/10.1103/PhysRevResearch.7.033064	*Physical Review Research, 7(3), 033064*
2024	**D. Hill, M. Bruno, H.P.M. Melo, Y. Takeuchi, V. Loreto. *Cities beyond proximity***, DOI: 10.1098/rsta.2024.0097	*Philosophical Transactions A*
2024	**M. Bruno, H.P. Monteiro Melo, B. Campanelli, V. Loreto. *A universal framework for inclusive 15-minute cities* (cover of the issue)** , DOI: 10.1038/s44284-024-00119-4	*Nature Cities*
2024	**F. Marzolla, M. Bruno, H.P.M. Melo, V. Loreto. *Compact 15-minute cities are greener***, arXiv:2409.01817	*arXiv preprint*
2023	**M. Bruno, D. Mazzilli, A. Patelli, T. Squartini, F. Saracco. *Inferring comparative advantage via entropy maximization***, DOI: 10.1088/2632-072X/ad1411	*Journal of Physics: Complexity*
2023	**Matteo Straccamore, Matteo Bruno, Bernardo Monechi, Vittorio Loreto. *Urban economic fitness and complexity from patent data***, Sci Rep 13, 3655 (2023)	*Scientific Reports*
2022	**Matteo Bruno, Renaud Lambiotte, Fabio Saracco. *Brexit and bots: characterizing the behaviour of automated accounts on Twitter during the UK election***, EPJ Data Sci. 11, 17 (2022)	*EPJ Data Science*
2021	**Nicolò Vallarano, Matteo Bruno, Emiliano Marchese, Giuseppe Trapani, Fabio Saracco, Tiziano Squartini, Giulio Cimini, Mario Zanon. *Fast and scalable likelihood maximization for Exponential Random Graph Models with local constraints***, Sci Rep 11, 15227 (2021)	*Scientific Reports*
2020	**Matteo Bruno, Fabio Saracco, Diego Garlaschelli, Claudio Tessone, Guido Caldarelli. *The ambiguity of nestedness under soft and hard constraints***, Sci Rep 10, 19903 (2020)	*Scientific Reports*
2019	**Matteo Bruno, Sandro Ferreira Sousa, Furkan Gürsoy, Matteo Serafino, Francesca Vianello, Ana Vranić, Marián Boguñá. *Community Detection in the Hyperbolic Space***, arXiv:1906.09082	*Preprint*
2018	**Matteo Bruno, Fabio Saracco, Tiziano Squartini, Marco Dueñas. *Colombian export capabilities: building the firms-products network***, Entropy 20.10 (2018): 785	*Entropy*